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Battery module and method for manufacturing the same

Abstract

The present invention relates to a battery module comprising: a plurality of battery cells disposed to overlap each other in a thickness direction thereof; a battery case configured to accommodate the battery cells and having a structure of which a lower portion is opened; and a heat dissipation member comprising a cover plate coupled to the lower portion of the battery case to support the battery cell and a heat dissipation body provided on one surface of the cover plate, on which the battery cell is supported, to dissipate heat generated in the battery cell, wherein the heat dissipation body comprises a first heat transfer material provided to be connected to a center of one surface of the cover plate in a longitudinal direction of the battery cell and a second heat transfer material provided on both portions of the first heat transfer material and having a structure aligned in a plurality of rows in the longitudinal direction of the battery cell, and the second heat transfer materials are aligned so that an interval therebetween is gradually narrowed from a center toward both ends of the battery cell to gradually improve heat dissipation performance from the center toward both the ends of the battery cell.

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Background/Summary

TECHNICAL FIELD
Cross-Reference to Related Application
(1) This application is a national stage entry under 35 U.S.C. § 371 of International Application No. PCT/KR2021/005531, filed on Apr. 30, 2021, published in Korean, which claims priority to Korean Patent Application No. 10-2020-0070473, filed on Jun. 10, 2020, the disclosures of which are hereby incorporated by reference herein in their entireties.
TECHNICAL FIELD
(2) The present invention relates to a battery module and a method for manufacturing the same, and more particularly, to a battery module, in which a battery cell increases in heat dissipation performance and decreases in temperature deviation, and a method for manufacturing the same.
BACKGROUND ART
(3) In general, secondary batteries refer to chargeable and dischargeable, unlike primary batteries that are not chargeable. The secondary batteries are being widely used for mobile phones, notebook computers, and camcorders, power storage device, electric vehicles, and the like.
(4) Such a secondary battery is classified into a can type secondary battery in which an electrode assembly is built in a metal can and a pouch type secondary battery in which an electrode assembly is built in a pouch. The pouch type secondary battery comprises an electrode assembly in which an electrode and a separator are alternately stacked, and a pouch accommodating the electrode assembly.
(5) As interests in the depletion of fossil fuels and environmental pollution increase, studies on hybrid vehicles and electric vehicles have been actively conducted in recent years, and a battery pack is mounted on each of the hybrid vehicles or electric vehicles.
(6) The battery pack comprises a battery module comprising a plurality of battery cells, and the plurality of battery cells are connected to each other in series or parallel to increase in capacity and

output.

(7) However, the above-described battery module generates more heat as the capacity and output increase, and thus, if the heat generated from the battery module is not smoothly released to the outside, deterioration, ignition, and explosion of the battery module may occur.

DISCLOSURE OF THE INVENTION

Technical Problem

(8) The present invention has been invented to solve the above problems, and according to the present invention, a heat dissipation structure of a battery module may be improved to smoothly discharge heat generated in the battery module to the outside, thereby improving heat dissipation performance of the battery module. Particularly, an object of the present invention is to provide a battery module that is capable of reducing a temperature deviation of the entire battery module and a method for manufacturing the same.

Technical Solution

(9) A battery module according to the present invention for achieving the above objects comprises: a plurality of battery cells disposed to overlap each other in a thickness direction thereof; a battery case configured to accommodate the battery cells and having a structure of which a lower portion is opened; and a heat dissipation member comprising a cover plate coupled to the lower portion of the battery case to support the battery cell and a heat dissipation body provided on one surface of the cover plate, on which the battery cell is supported, to dissipate heat generated in the battery cell, wherein the heat dissipation body comprises a first heat transfer material provided to be connected to a center of one surface of the cover plate in a longitudinal direction of the battery cell and a second heat transfer material provided on both portions of the first heat transfer material and having a structure aligned in a plurality of rows in the longitudinal direction of the battery cell, and the second heat transfer materials are aligned so that an interval therebetween is gradually narrowed from a center toward both ends of the battery cell to gradually improve heat dissipation performance from the center toward both the ends of the battery cell.

(10) The cover plate may comprise a first accommodation groove provided in a center of one surface thereof and having a structure connected in the longitudinal direction of the battery cell and second accommodation grooves provided in both sides of the first accommodation groove in a thickness direction of the battery cell and having a structure, in which an interval therebetween is gradually narrowed from the center toward both the ends of the battery cell, and

(11) the first heat transfer material is provided in the first accommodation groove, and the second heat transfer material is provided in the second accommodation groove.

(12) Since the first accommodation groove and the second accommodation groove are formed to be connected to each other, the first and second heat transfer materials may be integrally connected to each other.

(13) An insulating member having an insulating property may be provided on one surface of the cover plate except for the first and second accommodation grooves.

(14) A pair of heat dissipation pads, which reduces a temperature deviation between a center and both the ends of the battery cell by releasing the heat generated at both the ends of the battery cell may be provided on both ends of an inner surface of the battery case, respectively.

(15) A finishing pad that finishes a space between the pair of heat dissipation pads may be provided on the inner surface of the battery case between the pair of heat dissipation pads.

(16) The first accommodation groove may be provided to have a depth that gradually increases from the center toward both the ends of the battery cell, and the first heat transfer material provided in the first accommodation groove may be provided to have a thickness that gradually increases from the center toward both the ends of the battery cell.

(17) A depth may gradually increase from the second accommodation groove disposed at the center of the battery cell toward the second accommodation groove disposed at each of both the ends of the battery cell, and each of the second heat transfer materials provided in the second

accommodation groove may be provided to have a thickness that gradually increases from the center toward both the ends of the battery cell.

(18) A method for manufacturing a battery module according to the present invention comprises: a disposition step of disposing a plurality of battery cells to overlap each other in a thickness direction; an accommodation step of accommodating the plurality of overlapping battery cells in a battery case of which a lower portion is opened; a preparation step of preparing a heat dissipation member comprising a cover plate and a heat dissipation body provided on one surface of the cover plate, on which the battery cells are supported; and a coupling step of coupling the cover plate of the heat dissipation member to a lower portion of the battery case and supporting the heat dissipation body on the battery cells, wherein the preparation step comprises a preparation process of preparing the cover plate, a forming process of pressing one surface of the cover plate to form a first accommodation groove and a second accommodation groove, and an injection process of injecting a heat transfer solution into the first accommodation groove and the second accommodation groove to manufacture the heat dissipation body, wherein the heat dissipation body comprises a first heat transfer material formed in the first accommodation groove and a second heat transfer material formed in the second accommodation groove, in the forming process, the first accommodation groove is formed to be connected to a center of one surface of the cover plate in a longitudinal direction of the battery cell, and the second accommodation groove is provided in each of both portions of the first accommodation groove and is formed to have a structure aligned in a plurality of rows in the longitudinal direction of the battery cell, the second accommodation grooves are formed so that an interval between the second accommodation grooves is gradually narrowed from the center toward both the ends of the battery cell on one surface of the cover plate, and the second heat transfer materials provided in the second accommodation groove have a structure, in which an interval between the second heat transfer materials is gradually narrowed from the center toward both the ends of the battery cell.

(19) In the forming process, the first accommodation groove and the second accommodation groove may be formed to be connected to each other, and in the injection process, since the first accommodation groove and the second accommodation groove are connected to each other, the heat dissipation body, in which the first heat transfer material and the second heat transfer material are integrally connected to each other, may be manufactured.

(20) The method may further comprise an attachment process of attaching an insulating member having an insulating property to one surface of the cover plate except for the first accommodation groove and the second accommodation groove between the forming process and the injection process.

(21) The accommodation step may further comprise a process of respectively attaching heat dissipation pads, each of which has heat dissipation performance, to both ends of an inner surface of the battery case, which correspond to both the ends of the battery cell.

(22) The accommodation step may further comprise a process of attaching a finishing pad to the inner surface of the battery case between the heat dissipation pads.

Advantageous Effects

(23) The battery module according to the present invention may comprise the plurality of battery cells, the battery case, and the heat dissipation member comprising the cover plate and the heat dissipation body. The heat dissipation body may comprise the first heat transfer material and the second heat transfer materials, which are arranged in a plurality of rows. Due to the above-described characteristics, the heat generated in the battery cell may be smoothly released through the first and second heat transfer materials, and thus, the increase in temperature of the battery module may be significantly suppressed.

(24) Particularly, the first heat transfer materials, may be aligned so that an interval therebetween is gradually narrowed from the center toward both the ends of the battery cell. Due to the above-described characteristic, the heat dissipation performance may be gradually improved from the

center toward both the ends of the battery cell, and thus, the temperature deviation from the center to both the ends of the battery cell may be reduced to improve the performance of the battery module.
(25) In addition, in the battery module according to the present invention, the cover plate may comprise the first accommodation groove and the second accommodation groove, which are formed in a concave shape. The first heat transfer material may be provided in the first accommodation groove, and the second heat transfer material may be provided in the second accommodation groove. Due to the above-described characteristics, the first and second heat transfer materials may be effectively provided.
(26) In addition, in the battery module according to the present invention, the insulating member having the insulating property may be provided on one surface of the cover plate except for the first and second accommodation grooves. Due to the above-described characteristic, the occurrence of the short circuit due to the contact between the battery cell and the cover plate may be previously prevented to improve the safety.
(27) In addition, in the battery module of the present invention, the first accommodation groove may have the depth that gradually increases from the center to both the ends of the battery cell. Due to the above-described characteristic, the first heat transfer material provided in the first accommodation groove may have the height that gradually increases from the center to both the ends of the battery cell, and thus, the heat dissipation performance at both the ends of the battery cell when compared to the center of the battery cell may be improved, and as a result, the temperature deviation of the entire battery cell may be reduced.
(28) In addition, in the battery module of the present invention, the second accommodation groove may have the depth that gradually increases from the center to both the ends of the battery cell. Due to the above-described characteristic, the second heat transfer material provided in the second accommodation groove may have the height that gradually increases from the center to both the ends of the battery cell, and thus, the heat dissipation performance at both the ends of the battery cell when compared to the center of the battery cell may be improved, and as a result, the temperature deviation of the entire battery cell may be reduced.
(29) In addition, in the battery module according to the present invention, the pair of heat dissipation pads may be provided at both the ends of the inner surface of the battery case. Due to the above-described characteristic, the heat dissipation performance at both the ends of the upper portion of the battery cell may be significantly improved, and thus, the temperature deviation of the entire battery cell may be significantly reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a battery module according to a first embodiment of the present invention.
(2) FIG. **2** is a cross-sectional view of the battery module according to the first embodiment of the present invention.
(3) FIG. **3** is a perspective view illustrating a heat dissipation member of the battery module according to the first embodiment of the present invention.
(4) FIG. **4** is a plan view illustrating the heat dissipation member of the battery module according to the first embodiment of the present invention.
(5) FIG. **5** is a cross-sectional view taken along line A-A of FIG. **4**.
(6) FIG. **6** is a cross-sectional view taken along line B-B of FIG. **4**.
(7) FIG. **7** is a cross-sectional view taken along line C-C of FIG. **4**.
(8) FIG. **8** is a perspective view of a heat dissipation pad and a finishing pad in the battery module

according to the first embodiment of the present invention.
(9) FIG. **9** is a flowchart illustrating a method for manufacturing the battery module according to the first embodiment of the present invention.
(10) FIGS. **10** to **13** are flowcharts illustrating a preparation step of the method for manufacturing the battery module according to the first embodiment of the present invention.
(11) FIG. **14** is a cross-sectional view illustrating a heat dissipation member of a battery module according to a second embodiment of the present invention.
(12) FIG. **15** is a cross-sectional view illustrating a heat dissipation member of a battery module according to a third embodiment of the present invention.
MODE FOR CARRYING OUT THE INVENTION
(13) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings in such a manner that the technical idea of the present invention may easily be carried out by a person with ordinary skill in the art to which the invention pertains. The present invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. In the drawings, anything unnecessary for describing the present invention will be omitted for clarity, and also like reference numerals in the drawings denote like elements.
Battery Module According to First Embodiment of the Present Invention
(14) As illustrated in FIGS. **1** to **8**, in a battery module **100** according to a first embodiment of the present invention, a heat dissipation structure is improved to improve heat dissipation performance of the battery cell and reduce a temperature deviation of the entire battery cell. The battery module **100** comprises a plurality of battery cells **110** disposed to overlap each other in a thickness direction thereof, a battery case **120** accommodating the battery cells **110** and having a structure of which a lower portion is opened, and a heat dissipation member **130** coupled to the lower portion of the battery case **120** to release heat generated in the plurality of battery cells **110**, which are accommodated in the battery case **120**, to the outside.
(15) Here, in the battery module **100** according to the first embodiment of the present invention, the heat dissipation member is coupled to the lower portion of the battery case as one embodiment, but the heat dissipation member may be provided on an upper portion, a side portion, a front portion, or a rear portion of the battery case according to application of a product.
(16) Battery Cell
(17) The battery cell **110** comprises an electrode assembly, an electrode lead connected to the electrode assembly, and a pouch case accommodating the electrode assembly in a state in which a front end of the electrode lead is drawn out.
(18) The battery cell **110** having the above-described configuration is provided in plurality, which are disposed to overlap each other in the thickness direction, and the plurality of battery cells **110** arranged in the thickness direction have a structure, in which the plurality of battery cells **110** are connected in series or in parallel.
(19) Battery Case
(20) The battery case **120** is configured to accommodate the plurality of battery cells and has a rectangular box shape in which an opening is formed in a lower portion thereof. The plurality of battery cells **110** overlapping each other are accommodated in the battery case **120** through the opening.
(21) Each of the battery cells **110** is accommodated in the battery case **120** in a state in which the electrode lead faces an end in a longitudinal direction of the battery case **120** and is upright.
(22) Heat Dissipation Member
(23) The heat dissipation member **130** comprises a cover plate **131** supporting the battery cell **110** accommodated in the battery case **120** and a heat dissipation body **132** releasing heat generated in the battery cell **110** to the outside.
(24) The cover plate **131** may be coupled to the lower portion of the battery case **120** to finish the

lower portion of the battery case **120** and also supports the lower portion of the battery cell **110** accommodated in the battery case **120** to prevent the battery cell **110** from being drawn out to the outside.
(25) Particularly, the cover plate **131** is made of a heat dissipation material that is capable of smoothly releasing the heat of the battery cell, which is transferred from the heat dissipation body.
(26) The heat dissipation body **132** is provided on one surface of the cover plate **131**, on which the battery cell **110** is supported, to absorb the heat generated in the battery cell **110**, thereby releasing the heat to the outside. Thus, the heat of the battery cell **110** may be effectively dissipated.
(27) Particularly, the heat dissipation body **132** comprises a first heat transfer material **132***a* provided to be connected to a center of one surface of the cover plate **131** in a longitudinal direction (a left and right direction of the cover plate when viewed in FIG. **4**) of the battery cell and a second heat transfer material **132***b* provided at each of both portions (upper and lower portions of the first heat transfer material when viewed in FIG. **4**) of the first heat transfer material **132***a* and having a structure arranged in a plurality of rows in the longitudinal direction (the left and right direction of the cover plate when viewed in FIG. **4**) of the battery cell.
(28) Each of the first and second heat transfer materials **132***a* and **132***b* may be provided as a contact thermal interface material (TIM) and have an adhesive property. Particularly, at least one of heat dissipation grease, a thermal conductive adhesive, or a phase change material may be used as each of the first and second heat transfer materials **132***a* and **132***b*.
(29) The heat dissipation body **132** having the above-described configuration may be differently applied to a center of the battery cell **110**, at which a relatively large amount of heat is generated, and both overlapping portions of the battery cell **110**, at which a relatively small amount of heat when compared to that of the center of the battery cell **110** is generated, and thus, the heat dissipation performance of the battery cell **110** may be improved, and also, the temperature deviation between the center and both the portions of the battery cell **110** may be reduced.
(30) In the battery cell **110**, relatively high-temperature heat is generated at both ends rather than a central portion of the battery cell **110** due to high resistance by the electrode lead. The present invention has the arrangement structure of the second heat transfer materials **132***b* for reducing the temperature deviation of the battery cell **110** as described above.
(31) That is, the second heat transfer materials **132***b* have a structure aligned in a plurality of rows so that an interval therebetween is gradually narrowed from an outer central point (a point at which the battery cell is bisected in the longitudinal direction) toward both ends (both end points of the battery cell in the longitudinal direction). Thus, more second heat transfer materials **132***b* are arranged at both the ends of the battery cell, which generates high-temperature heat are arranged to significantly improve the heat dissipation performance, and less second heat transfer materials **132***b* are arranged at the central portion of the battery cell, which generate low-temperature heat to slightly improve the heat dissipation performance. As a result, the temperature deviation may be significantly reduced due to a difference in heat dissipation performance between the central portion and both the ends of the battery cell.
(32) The heat dissipation member **130** having the above configuration may significantly improve the heat dissipation performance of the plurality of battery cells **110** accommodated in the battery case **120**, and in particular, may reduce the temperature deviation of the plurality of entire battery cells to improve performance of the battery cells.
(33) In the heat dissipation member **130**, the cover plate **131** comprises a first accommodation groove **131***a* provided in a center of one surface thereof and having a structure connected in a longitudinal direction of the battery cell **110** and second accommodation grooves **131***b* respectively provided in both sides of the first accommodation groove **131***a* in a thickness direction of the battery cell **110** and having a structure, in which an interval therebetween is gradually narrowed from the center toward both ends of the battery cell **110**.
(34) Here, the first heat transfer material **132***a* is provided in the first accommodation groove **131***a*,

and the second heat transfer material **132***b* is provided in the second accommodation groove **131***b*.
(35) The first accommodation groove **131***a* and the second accommodation groove **131***b* have a structure formed in a concave shape on a top surface of the cover plate **131** when viewed in FIG. **3**.
(36) Also, the first heat transfer material **132***a* and the second heat transfer materials **132***b* are made of the same material.
(37) The heat dissipation member **130** having the above-described configuration forms the first accommodation groove **131***a* and the second accommodation groove **131***b* in one surface of the cover plate **131** to easily form the first heat transfer material **132***a* and the second heat transfer materials **132***b*.
(38) The first accommodation groove **131***a* and the second accommodation groove **131***b* are formed to be connected to each other. Thus, the first and second heat transfer materials **132***a* and **132***b* provided in the first accommodation groove **131***a* and the second accommodation groove **131***b* may be integrally connected to each other, and as a result, heat generated in the battery cell **110** may be dispersed throughout the first and second heat transfer materials **132***a* and **132***b* to improve the heat dissipation performance.
(39) An insulating member **133** having an insulating property is provided on one surface of the cover plate **131** except for the first and second accommodation grooves **131***a* and **131***b*. That is, the insulating member **133** insulates the battery cell **110** and the cover plate **131** from each other to prevent a short circuit that may occur when the battery cell **110** and the cover plate **131** are in contact with each other.
(40) The insulating member **133** has an adhesive tape shape and is attached to one surface of the cover plate **131**. Therefore, convenience of use may be improved.
(41) Particularly, a coating layer is further provided between an end of the insulating member **133** and the cover plate **131** to prevent the heat transfer material from being introduced between the end of the insulating member **133** and the cover plate. Particularly, the coating layer may prevent the end of the insulating member **133** from being separated the cover plate **131**.
(42) The battery module **100** according to the first embodiment of the present invention further comprises a heat dissipation pad **140**.
(43) Heat Dissipation Pad
(44) The heat dissipation pad **140** is configured to reduce a temperature deviation between the upper central portion and both the ends of the battery cell
(45) That is, both the ends of the battery cell generate heat having a temperature higher than that of the center of the battery cell because the electrode leads are connected, and thus, the heat dissipation pads **140** may be further provided to reduce the temperature deviation between the center of the top surface and both the ends of the battery cell.
(46) The heat dissipation pads **140** are attached to both sides of an inner surface of the battery case **120**, respectively, and both ends of the top surface of the battery cell **110** accommodated in the battery case **120** are supported to release heat through both the ends of the top surface of the battery cell **110**. Thus, the temperature deviation between the center and both the ends of the battery cell may be significantly reduced through the increase in heat dissipation performance at both the ends of the battery cell.
(47) A finishing pad **150** may be further provided to constantly maintain an interval between the heat dissipation pads **140**, which are attached to both sides of the inner surface of the battery case **120**, respectively.
(48) Finishing Pad
(49) The finishing pad **150** is attached to the inner surface of the battery case **120** between the pair of heat dissipation pads **140** to constantly maintain the interval between the pair of heat dissipation pads **140**. Particularly, the finishing pad **150** may finish a space between the pair of heat dissipation pads **140** to prevent the battery cell **110** from being deformed because the battery cell **110** is inserted into the space between the pair of heat dissipation pads **140**.

(50) Hereinafter, a method for manufacturing the battery module according to the first embodiment of the present invention will be described.

Method for Manufacturing Battery Module According to First Embodiment of the Present Invention

(51) As illustrated in FIGS. **9** to **13**, a method for manufacturing the battery module according to the first embodiment of the present invention comprises a disposition step, an accommodation step, a preparation step, and a coupling step.

(52) Disposition Step

(53) In the disposition step, a plurality of battery cells **110** are prepared, the plurality of prepared battery cells **110** are disposed to overlap each other in a thickness direction, and the plurality of battery cells **110**, which are disposed to overlap each other, are connected to each other in series or parallel to be in contact with each other.

(54) Accommodation Step

(55) In the accommodating step, the plurality of battery cells **110** overlapping each other are accommodated in the battery case **120** having an opened lower portion. Here, each of the battery cells **110** is accommodated in a state in which the electrode lead faces an end in a longitudinal direction of the battery case **120** and is upright.

(56) The accommodation step further comprises a step of attaching the heat dissipation pad **140** to both ends of an inner surface of the battery case **120**, which correspond to both ends of a top surface of the battery cell **110**, respectively, and the heat dissipation pad **140** improves heat dissipation performance at both the ends of the top surface of the battery cell **110**.

(57) In addition, the accommodation step further comprises a step of attaching a finishing pad **150** to the inner surface of the battery case **120** between the heat dissipation pads **140**, and the finishing pad **150** constantly maintains an interval between the pair of heat dissipation pads **140** to finish a space between the pair of heat dissipation pads **140**.

(58) Preparation Step

(59) The preparation steps comprises a preparation process of preparing the cover plate **131**, a forming process of forming an accommodation groove for providing a heat dissipation body in the cover plate **131**, and an injection process of manufacturing a heat dissipation body **132** through the accommodation groove formed in a cover plate **131**.

(60) Referring to FIG. **10**, in the preparation process, the cover plate **131** having a size and shape corresponding to an opened lower portion of the battery case **120** is prepared.

(61) Referring to FIG. **11**, in the forming process, a first accommodation groove **131***a* and a second accommodation groove **131***b*, each of which is formed in a concave shape by pressing one surface (a top surface of the cover plate when viewed in FIG. **11**) of the cover plate **131** by using a press, are formed.

(62) Here, the first accommodation groove **131***a* is formed to be connected to a center of one surface of the cover plate **131** in a longitudinal direction of the battery cell, and the second accommodation groove **131***b* is provided in both portions of the first accommodation groove **131***a* and having a structure aligned in a plurality of rows in the longitudinal direction of the battery cell **110**.

(63) Particularly, the second accommodation grooves **131***b* are formed in one surface of the cover plate **131** so that an interval between the second accommodation grooves **131***b* is gradually narrowed from a center toward both ends of the battery cell **110**. Thus, the second heat transfer materials **132***b* provided in the second accommodation groove **131***b* may be manufactured to have a structure in which an interval between the second heat transfer materials **132***b* is gradually narrowed from the center toward both the ends of the battery cell **110**.

(64) A heat dissipating member **130** provided with a heat dissipation body **132** provided on one surface of the cover plate **131**, on which the battery cell **110** is supported, is prepared.

(65) In the forming process, the first accommodation groove **131***a* and the second accommodation

groove **131***b* are formed to be connected to each other. Thus, even if a heat transfer solution is injected into the first accommodation groove **131***a* or the second accommodation groove **131***b*, the heat transfer solution may be injected up to other accommodation grooves, and as a result, operation efficiency may be improved. Particularly, since the first and second heat transfer materials **132***a* and **132***b* manufactured in the first accommodation groove **131***a* and the second accommodation groove **131***b* may be manufactured to be integrally connected to each other, thereby improving heat dissipation performance.

(66) An attachment process of attaching an insulating member having an insulating property to one surface of the cover plate **131** except for the first accommodation groove **131***a* and the second accommodation groove **131***b* is performed between the forming process and the injection process.

(67) In the attachment process, referring to FIG. **12**, the insulating member **133** having adhesive force is attached to one surface of the cover plate **131**. Here, in order to prevent an end of the insulating member **133** from being separated, a coating solution is applied between an end of the insulating member **133** and the cover plate **131** to prepare a coating layer.

(68) Referring to FIG. **13**, in the injection process, a heat transfer solution **132***c* is injected into the first accommodation groove **131***a* and the second accommodation groove **131***b* to manufacture a heat dissipation body **132** comprising the first heat transfer material **132***a* and the second heat transfer material **132***b*. That is, as the heat transfer solution **132***c* is applied to the first accommodation groove **131***a* and solidified, the first heat transfer material **132***a* is prepared, and as the heat transfer solution **132***c* is applied to the second accommodation groove **131***b* and solidified, the second heat transfer material **132***b* is prepared. Here, since the first accommodation groove **131***a* and the second accommodation groove **131***b* are connected to each other, the first heat transfer material **132***a* and the second heat transfer material **132***b* may be manufactured to be integrally connected to each other.

(69) Therefore, when the method for manufacturing the battery module according to the first embodiment of the present invention is completed, the finished battery module **100** illustrated in FIG. **2** may be manufactured.

(70) Hereinafter, in descriptions of another embodiment of the present invention, constituents having the same function as the above-mentioned embodiment have been given the same reference numeral in the drawings, and thus duplicated description will be omitted.

Battery Module According to Second Embodiment of the Present Invention

(71) As shown in FIG. **14**, a battery module **100** according to a second embodiment of the present invention comprise a first accommodation groove **131***a* provided in a center of one surface thereof and having a structure connected in a longitudinal direction of the battery cell **110** and second accommodation grooves **131***b* provided in both sides of the first accommodation groove **131***a* in a thickness direction of the battery cell **110** and having a structure, in which an interval therebetween is gradually narrowed from the center toward both ends of the battery cell **110**.

(72) Here, the first accommodation groove **131***a* is provided to gradually increase in depth from the center to both the ends of the battery cell **110**, and thus, the first heat transfer material **132***a* provided in the first accommodation groove **131***a* has a structure in which a height thereof gradually increases from the center to both the ends of the battery cell **110**.

(73) Therefore, in the battery module **100** according to the second embodiment of the present invention, heat dissipation performance may be applied differently from the middle to both the ends of the battery cell **110** disposed at the center, and thus, the battery cell disposed at the center of the battery case may be reduced in temperature deviation.

Battery Module According to Third Embodiment of the Present Invention

(74) As shown in FIG. **15**, a battery module **100** according to a third embodiment of the present invention comprise a first accommodation groove **131***a* provided in a center of one surface thereof and having a structure connected in a longitudinal direction of the battery cell **110** and second accommodation grooves **131***b* provided in both sides of the first accommodation groove **131***a* in a

thickness direction of the battery cell **110** and having a structure, in which an interval therebetween is gradually narrowed from the center toward both ends of the battery cell **110**.
(75) Here, the second accommodation groove **131***b* is provided to gradually increase in depth from the center to both the ends of the battery cell **110**, and thus, the second heat transfer material **132***b* provided in the second accommodation groove **131***b* has a structure in which a height thereof gradually increases from the center to both the ends of the battery cell **110**.
(76) Therefore, in the battery module **100** according to the second embodiment of the present invention, heat dissipation performance may be more precisely adjusted from the middle to both the ends of the battery cell **110** disposed at each of both the ends of the battery case, and thus, the battery cell disposed at each of both the ends of the battery case may be reduced in temperature deviation.
(77) Accordingly, the scope of the present invention is defined by the appended claims more than the foregoing description and the exemplary embodiments described therein. Various modifications made within the meaning of an equivalent of the claims of the invention and within the claims are to be regarded to be in the scope of the present invention.
DESCRIPTION OF THE SYMBOLS
(78) **100**: Battery module **110**: Battery cell **120**: Battery case **130**: Heat dissipation member **131**: Cover plate **131***a*: First accommodation groove **131***b*: Second accommodation groove **132**: Heat dissipation body **132***a*: First heat transfer material **132***b*: second heat transfer material **133**: Insulating member **140**: Heat dissipation pad **150**: Finishing pad

Claims

1. A battery module comprising: a plurality of battery cells overlapping each other in a thickness direction thereof; a battery case accommodating the battery cells therein and having a structure in which a lower portion is opened; and a heat dissipation member comprising a cover plate coupled to the lower portion of the battery case and supporting the battery cells and a heat dissipation body disposed on a first surface of the cover plate, the battery cells being supported on the heat dissipation member, the heat dissipation member being configured to dissipate heat generated in the battery cells, wherein the heat dissipation body comprises a first heat transfer material connected to a center of a first surface of the cover plate in a longitudinal direction of the battery cells and a second heat transfer material extending from opposite sides of the first heat transfer material and shaped in a plurality of adjacent rows in the longitudinal direction of the battery cells, and the adjacent rows of the second heat transfer material are aligned so that intervals therebetween are gradually narrowed from a center of the battery cells toward two opposite ends of the battery cells, the second heat transfer material being configured to improve heat dissipation performance in the longitudinal direction from the center toward the two opposite ends of the battery cells.
2. The battery module of claim 1, wherein the cover plate comprises a first accommodation groove disposed in a center of the first surface thereof and extending in the longitudinal direction of the battery cells and second accommodation grooves extending from opposite sides of the first accommodation groove in a thickness direction of the battery cells, the second accommodation grooves defining intervals therebetween that are gradually narrowed from the center toward the two opposite ends of the battery cells, and the first heat transfer material is disposed in the first accommodation groove, and the second heat transfer material is disposed in the second accommodation grooves.
3. The battery module of claim 2, wherein, the first accommodation groove and the second accommodation groove are connected to each other, and the first and second heat transfer materials are integrally connected to each other.
4. The battery module of claim 2, further comprising an insulating member having an insulating property disposed on the first surface of the cover plate at locations outside of the first and second

accommodation grooves.

5. The battery module of claim 1, further comprising a pair of heat dissipation pads, the heat dissipation pads being configured to reduce a temperature deviation between the center and the two opposite ends of the battery cells by releasing heat generated at the opposite ends of the battery cells, the pair of heat dissipation pads being disposed on opposite ends of an inner surface of the battery case, respectively.

6. The battery module of claim 5, further comprising a finishing pad that is disposed within a space between the pair of heat dissipation pads, the finishing pad being is disposed on the inner surface of the battery case between the pair of heat dissipation pads.

7. The battery module of claim 2, wherein the first accommodation groove is has a depth that gradually increases from the center toward the two opposite ends of the battery cells, and the first heat transfer material provided in the first accommodation groove has a thickness that gradually increases from the center toward the two opposite ends of the battery cells.

8. The battery module of claim 2, wherein the second accommodation grooves have depths that gradually increase from the center of the battery cells toward the two opposite ends of the battery cells, and the rows of the second heat transfer material have thicknesses that gradually increase from the center toward the two opposite ends of the battery cells.

9. A method for manufacturing a battery module, the method comprising: arranging a plurality of battery cells to overlap each other in a thickness direction; accommodating the plurality of battery cells within a battery case of which a lower portion is opened; preparing a heat dissipation member comprising a cover plate and a heat dissipation body disposed on a first surface of the cover plate, the battery cells being supported on the heat dissipation member; and coupling the cover plate of the heat dissipation member to the lower portion of the battery case and positioning the heat dissipation body on the battery cells, wherein the preparing comprises a preparation process of preparing the cover plate, a forming process of pressing the first surface of the cover plate to form a first accommodation groove and second accommodation grooves, and an injection process of injecting a heat transfer solution into the first accommodation groove and the second accommodation grooves to manufacture the heat dissipation body, wherein the heat dissipation body comprises a first heat transfer material formed in the first accommodation groove and a second heat transfer material formed in the second accommodation grooves, in the forming process, the first accommodation groove is formed from a center of the first surface of the cover plate and extends in a longitudinal direction of the battery cells, and the second accommodation grooves extend from opposite sides of the first accommodation groove and are shaped in a plurality of adjacent rows in the longitudinal direction of the battery cells, the second accommodation grooves define intervals therebetween that are gradually narrowed from a center of the battery cells toward two opposite ends of the battery cells on one the first surface of the cover plate, and adjacent rows of the second heat transfer material define intervals therebetween that are gradually narrowed from the center toward the two opposite ends of the battery cells.

10. The method of claim 9, wherein the first accommodation groove and the second accommodation groove are connected to each other, and during the injection process, the first heat transfer material and the second heat transfer material are formed integrally connected to each other.

11. The method of claim 9, further comprising attaching an insulating member having an insulating property to the first surface of the cover plate at locations outside of the first accommodation groove and the second accommodation grooves, the attaching being between the forming process and the injection process.

12. The method of claim 9, wherein the accommodating further comprises attaching heat dissipation pads, each of which has a heat dissipation property, to opposite ends of an inner surface of the battery case that are adjacent to the two opposite ends of the battery cells, respectively.

13. The method of claim 12, wherein the accommodating further comprises attaching a finishing pad to the inner surface of the battery case between the heat dissipation pads.